

Determining Access Permissions

Access permissions are applied when you try to access the data locally through file system direct, there is another type of permissions (sharing Permissions) are applied when you try to access the data via the network with combination between access permissions and the most restrictive combinations take effect.

Permission Classes

Users are categorized into three unique classes for maintaining file security through access rights.

Permission Class	Description
User (u)	The owner of file or directory. Usually, the file creator is its owner.
Group (g)	A set of users that need identical access on files and directories that they share. Group information is maintained in the /etc/group file and user are assigned to groups according to shared file access needs.
Others (o)	All other users on the system except for the owner and group members. Also called public.

s

Permission Types

Permissions control what actions can be performed on a file or directory and by whom. There are three types of permissions.

Perm Type	Symbol	File	Directory
Read	r	Lets us view file contents and copy them to another file.	Lets us view contents with the ll command.
Write	w	Allows us to modify file contents.	Allows us to create, remove, or rename files and sub-directories.
Execute	x	Lets us execute a file.	Allows us to cd into the directory.

We can simplify the permissions in what command can you apply

Perm Type	File	Directory
Read	Cat, tail, head, less, more. And Open the file with one of editors but in read only mode (nano, vi and vim)	Ls (with all options) ll
Write	Edit in text file (open the file and modify then save the modification)	Mkdir, touch, rm, cp, mv
Execute	Executable file	cd

If a read, write, or execute permission is not desired, the – sign is used to represent its absence.

Permission Modes

A permission mode is used to add, revoke, or assign a permission type to a permission class. Table 3-5 shows various permission modes.

Permission Mode	Description
Add (+)	Allocates permissions.
Revoke (-)	Removes permissions.
Assign (=)	Allocates permissions to owner, group members, and public at once.

Examples

1. Add the execute permission for the owner and show the output of ll to verify:

```
$ chmod u+x file1 -v
mode of 'file1' changed from 0444 (r--r--r--) to 0544 (r-xr--r--)
$ ll file1
-r-xr--r--. 1 user1 user1 0 Nov 30 12:30 file1
```

2. Add the write permission for group members and public and verify:

```
$ chmod go+w file1
$ ll file1
-r-xrw-rw-. 1 user1 user1 0 Nov 30 12:30 file1
```

3. Remove the write permission for the public and confirm:

```
$ chmod o-w file1
$ ll file1
-r-xrw-r--. 1 user1 user1 0 Nov 30 12:30 file1
```

4. Assign read, write, and execute permissions to all three user categories and validate:

```
$ chmod a=rwx file1
$ ll file1
-rwxrwxrwx. 1 user1 user1 0 Nov 30 12:30 file1
```

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